DS2002 - Midterm

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Data Cleaning

Walmart Transaction: 25,562 rows x 8 columns

Issues:

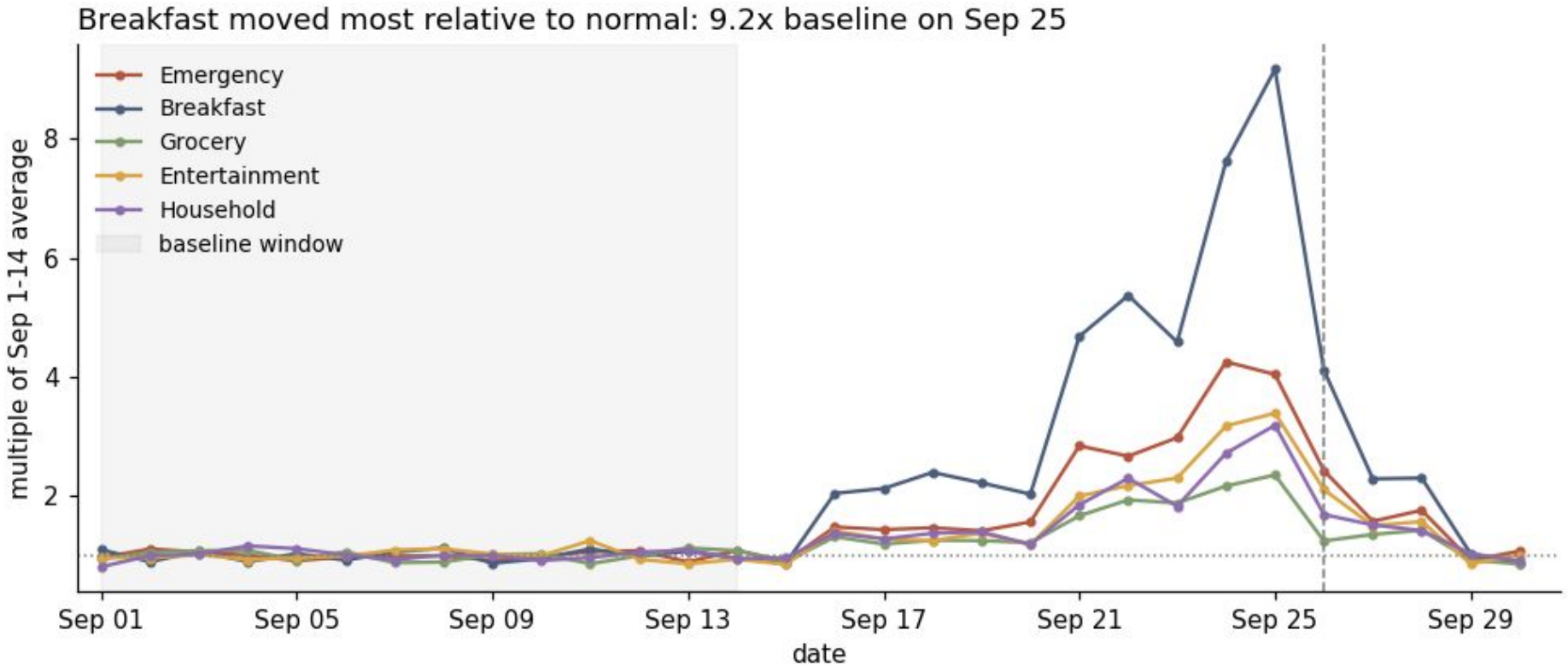
- Store_id, 60 distinct values for 12 stores
- timestamp, 5 different format 886 are blank, NULL, N/A, NaN, or nan
- unit_price, irregular data (text variable or start with “$” or “ “)
- SKUs - 45 SKU for 20 products
- category - 37 unique spelling for 10 categories, 130 negative quantities and 30 duplicate rows

Data Cleaning

Result:

- transactions: 25562 -> 24521 (1041 removed)
- stores: 14 -> 12
- date range: 2024-09-01 06:01:15 - 2024-09-30 22:57:47
- SKUs: 45 -> 20, 1 category per product

Question 1 - Demand Surge



Q1 - Results

Sep 1-14 baseline (about 580 transaction and 1,100 units a day):

- Demand rose 1.4x on Sep 16, 2.4x on Sep 21, peak on Sep 24-25 3.6x

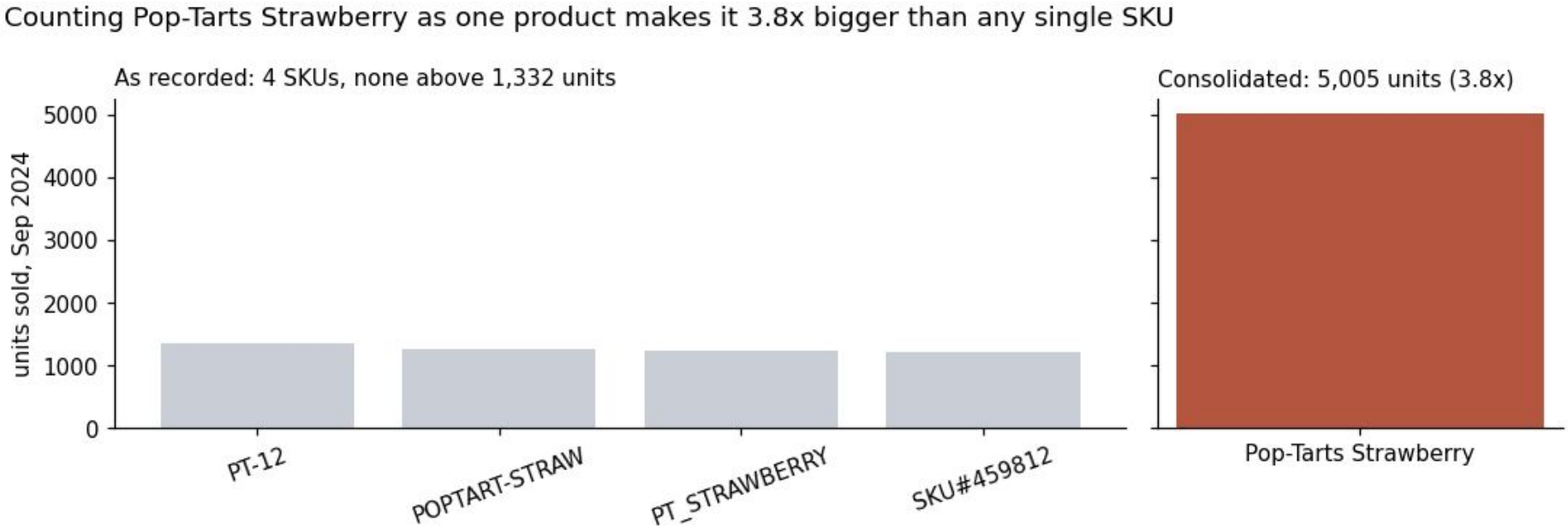
Demand fell on landfall day and return to normal level on Sep 29

Emergency supplies biggest in absolute value, breakfast in relative value (9.2x)

Between Sep 19-25:

- Breakfast (strawberry Pop-Tarts rose the most): +410% (74 to 377 units)
- Emergency supplies: +182% (357 to 1,006 units)
- Electronics: +152%
- Alcohol: +125%
- Health: +124%
- Entertainment: +123%
- Household: +106%
- Grocery: +78%
- Baby: +57%
- Snacks: +24%

Question 2 - SKU consolidation



Q2 - results

Sku before consolidation:

- WTR_24PK: 1758
- WATER24PK: 1657
- WTR-24: 1622
- BATT-100: 1432
- BAT-AA4: 1427

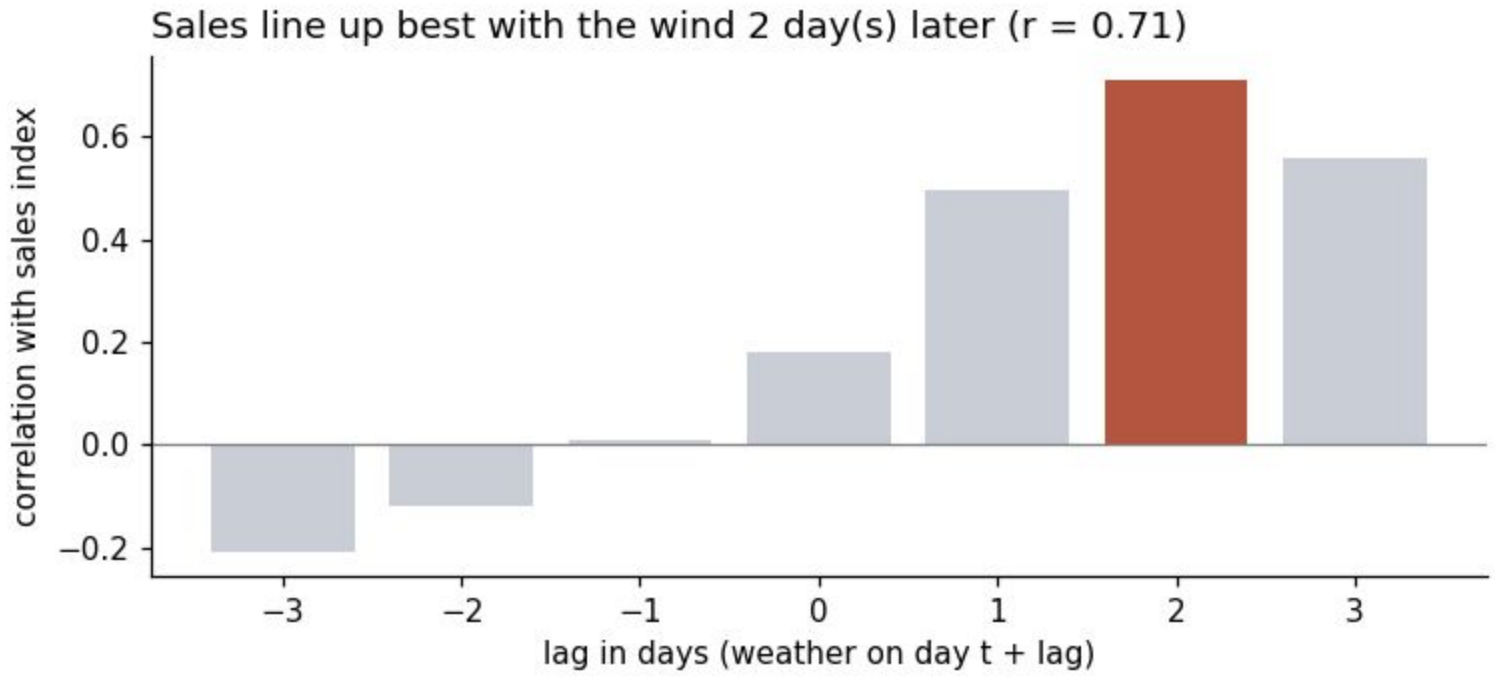
SKU after consolidation:

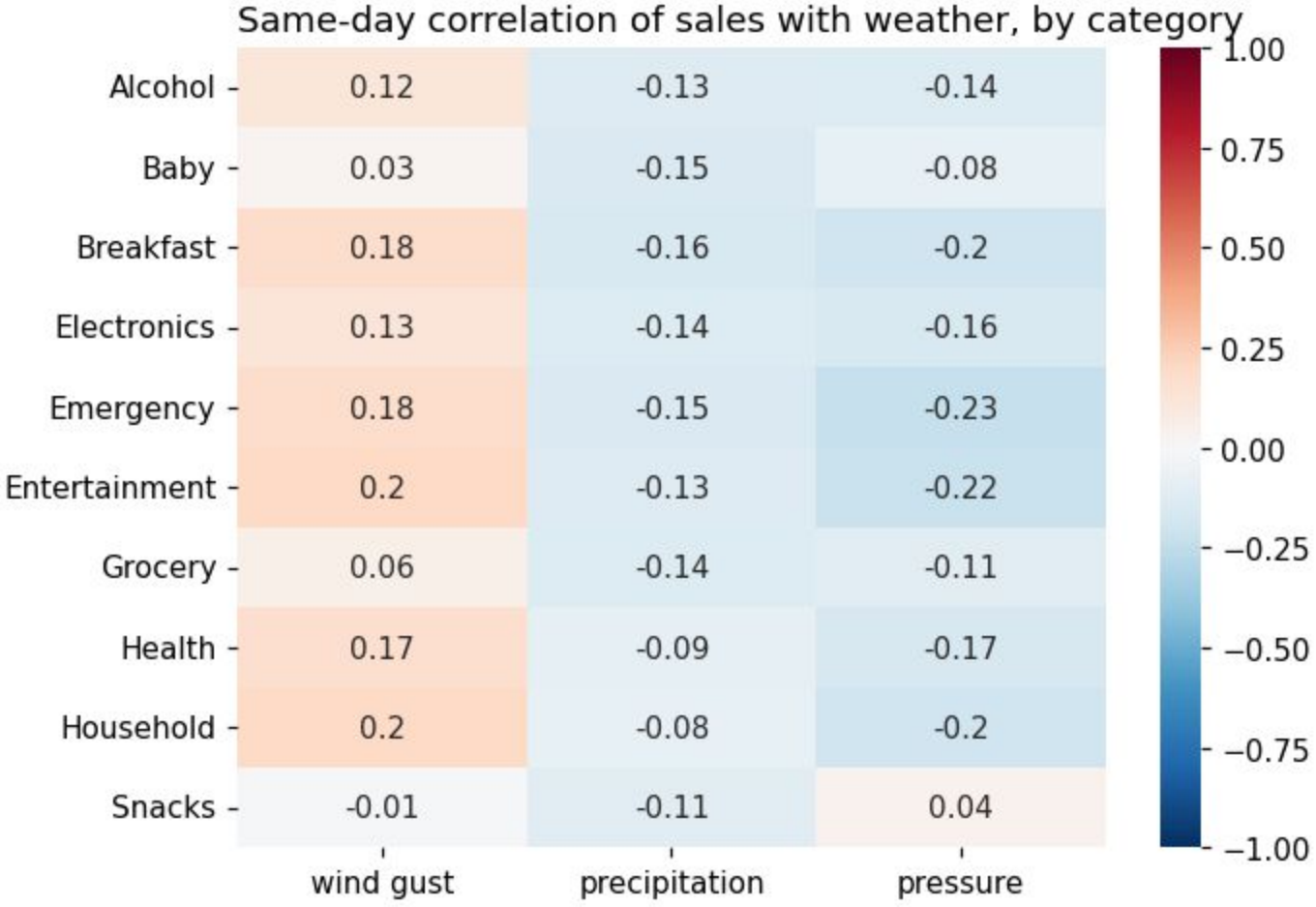
- Bottled Water 24-Pack: 5037
- Pop-Tarts Strawberry: 5005
- AA Batteries 4-Pack: 4154
- LED Flashlight: 2814
- Emergency Candles: 2546

Without consolidation we would miss the surge in Pop-Tarts (5,005 units, 3.8x any single SKU, #2 product)

Question 3 - Weather Correlation




Sales line up best with the wind 2 day(s) later (r = 0.71)
correlation with sales index
0.6
0.4
0.2
0.0
−0.2
−3
−2
−1
0
1
2
3
lag in days (weather on day t + lag)


Same-day correlation of sales with weather, by category
Alcohol
Baby
Breakfast
Electronics
Emergency
Entertainment
Grocery
Health
Household
Snacks
0.12
-0.13
-0.14
0.03
-0.15
-0.08
0.18
-0.16
-0.2
0.13
-0.14
-0.16
0.18
-0.15
-0.23
0.2
-0.13
-0.22
0.06
-0.14
-0.11
0.17
-0.09
-0.17
0.2
-0.08
-0.2
-0.01
-0.11
0.04
wind gust
precipitation
pressure
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Q3 - Results

Same-day weather does not explain sales across all stores the correlation is: r = 0.18 (wind gust), r = -0.16 (rain), r = -0.23 (pressure)

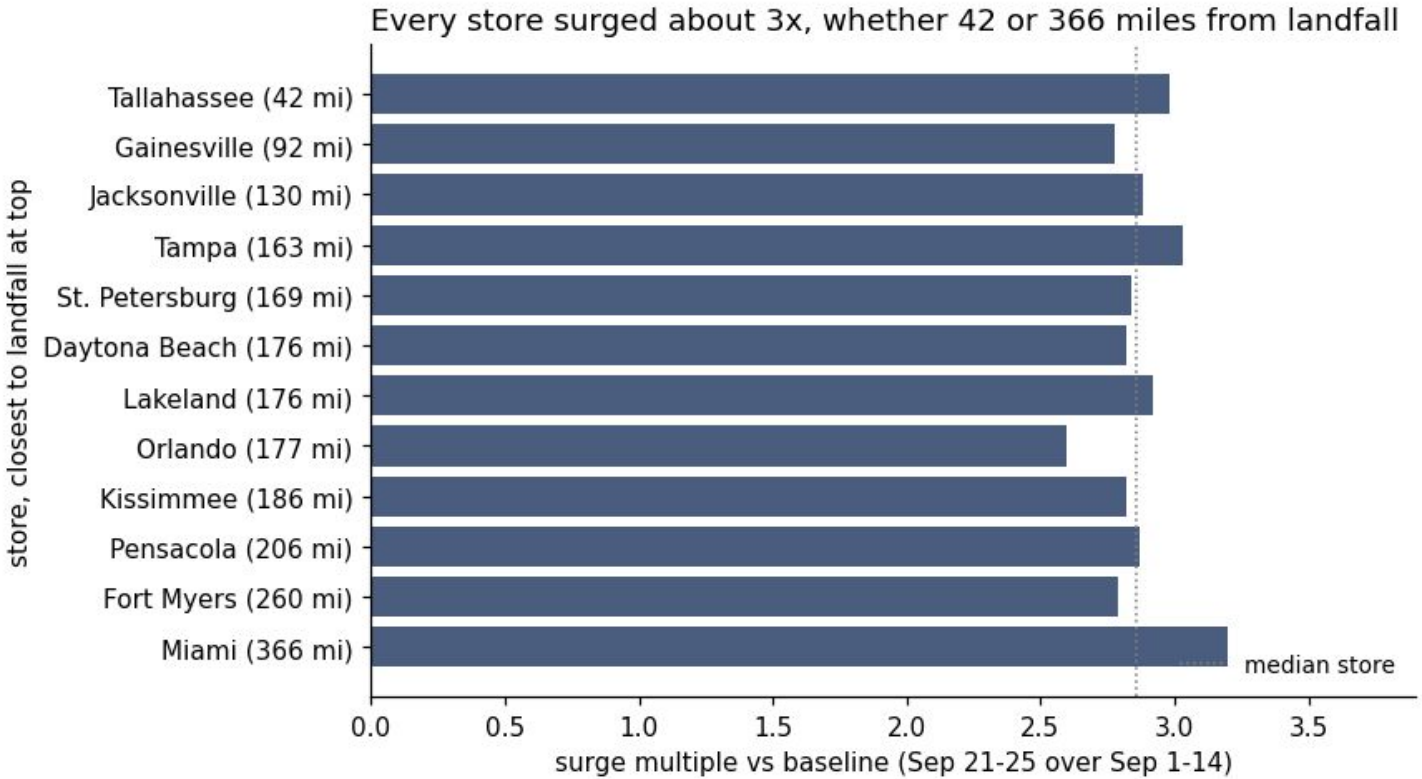
Weather is a lagging indicator when compared to two day later, wind gust r = 0.71, buying peaked Sep 24-25 when gust was 20-30 mph while strongest 58 mph appeared on Sep 26-27 when sales was falling

In almost all weather variable 2-day lag is the best indicator:

- Wind - +2 day-lag for every category expect Baby (+3), (0.49 to 0.69)
- Pressure - +2 day-lag, negative correlation (-0.69 to -0.60)
- Precipitation - weakest signal (-0.17 to -0.30) not a useful predictor

Snack and Baby category had the weakest response. There is a detectable lag sale rises two days before the worst wind and lowest pressure. Direct correlation between supplies needed for the storm (emergency supplies).

Question 4 - Stores and geography



Q4 - results

There is no distance gradient, all store surged between 2.6x and 3.2x, correlation between distance and surge -0.08

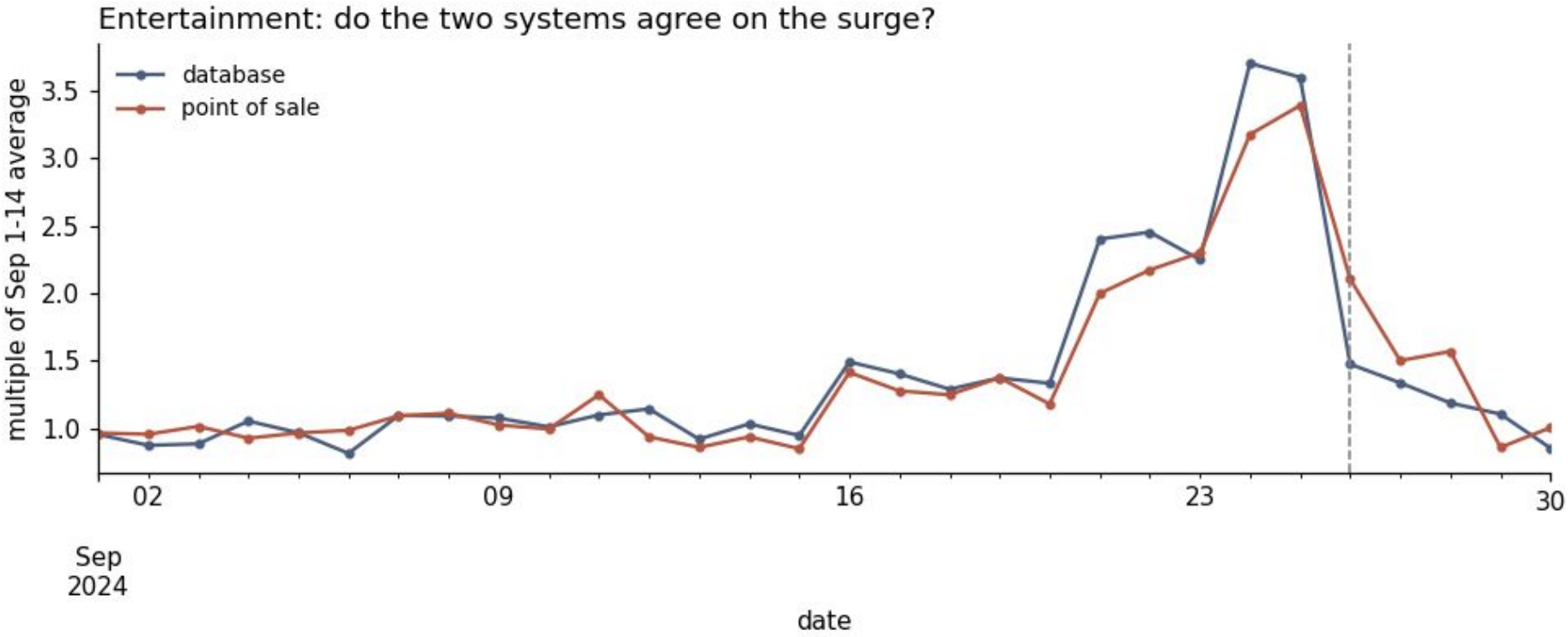
Tallahassee (42 miles from landfall): 2.98x surge

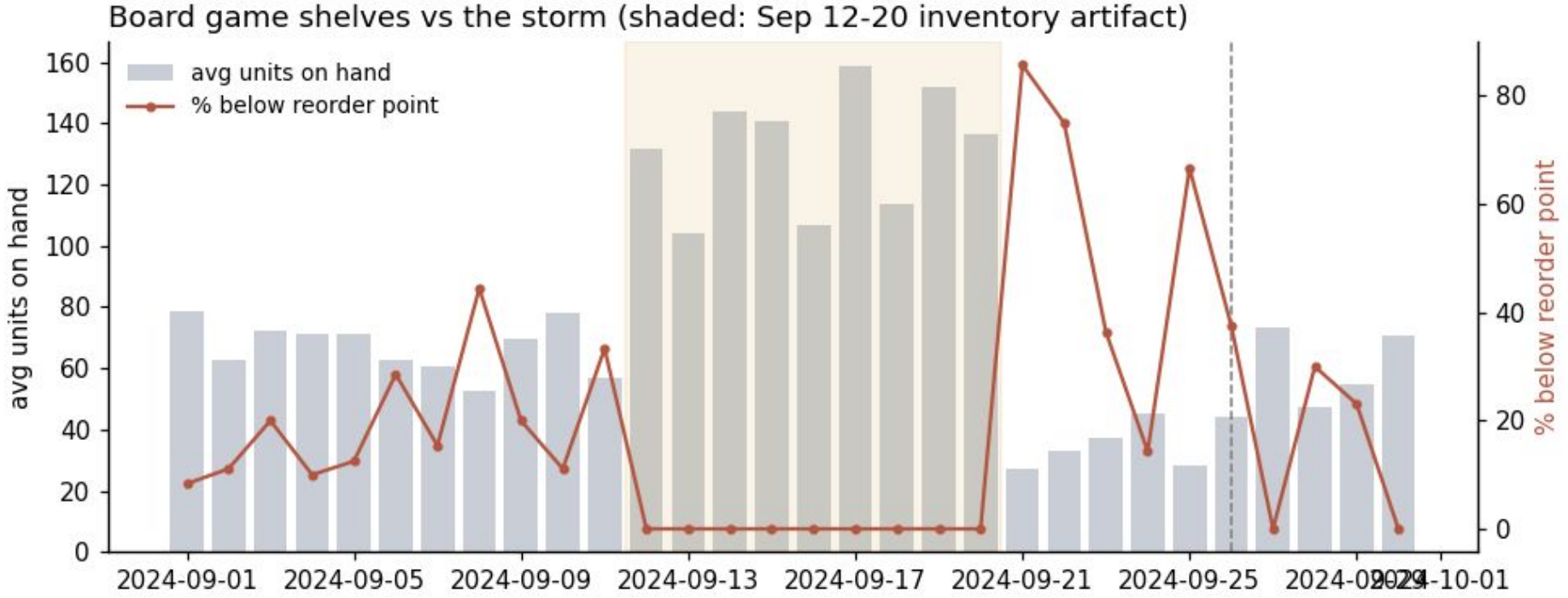
Miami (366 miles from landfall): 3.20x surge

Orlando: 2.60x surge (lowest surge)

Each store had ~2,000 transaction, differences not caused by small-sample noise

Question 5 - Signal or artifact




Board game shelves vs the storm (shaded: Sep 12-20 inventory artifact)
avg units on hand
% below reorder point
avg units on hand
% below reorder point
0
20
40
60
80
100
120
140
160
0
20
40
60
80
2024-09-01
2024-09-05
2024-09-09
2024-09-13
2024-09-17
2024-09-21
2024-09-25
2024-10-01

Q5 - results

Board game - signal not artifact

1. No category lumping - Entertainment holds only two products: Monopoly and playing cards. Each product accounts for about half of the surge units (49% and 51%), with the “+250%” figure matching the category’s single-day peak not the 7-day average.
2. Another system agrees - Database’s Entertainment totals follow the point-of-sale curve almost exactly (0.96 correlation)
3. Shelves emptied when sales surged - Average board game stock around 55-80 units a day until Sep 11. Sep 12-20 saw a jump to 105-160 units with zero SKUs below reorder point (artifact). Fell to 27-45 units on Sep 21-26, 86% of board game SKUs were below reorder point on Sep 21 (75% on Sep 22, 67% on Sep 25). Recovered to 73 units on Sep 27 after landfall showing real demand spike.

Recommendations

- Focus on forecast, not weather - demand started to climb 10 days before the hurricane, peaking 1-2 days before landfall. Storm stock should be in stores at least 5 days before landfall as half of the sampled SKUs were already below reorder point 5 days out
- Stock every store in Florida not just the path - Surge was 3x everywhere allocating orders based on distance to forecast track would have ignored stores like Miami and Tampa
- Plan Pop-Tarts as a storm item - at 9.2x baseline it moved more than water or flashlights
- FIx the data before the next storm - Standardize store ids and SKUs at the source, audit the Sep 12-20 inventory jump, do not use database category split until it matches point-of-sale data.
- Stock extra board games - Monopoly rose +157% in the week before landfall, database agrees with point-of-sale (r = 0.96), shelves ran below reorder point on Sep 21-26, plan for about 2.5x normal volume and have stock in stores about 5 days before landfall.